Traffic Congestion Prediction Using Machine Learning

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***Abstract*— Due to the world population growing rapidly over time, the number of personal and local vehicles are increasing which is one of the main causes of high traffic on the roads. For high traffic, the average speed of vehicles is decreasing which is known as traffic congestion. It is a very common and alarming problem in today's world. Due to traffic congestion, civilians are facing different problems in this 21st century. Time is a precious thing and traffic congestion is killing the most precious times of our lives. In this paper, the authors aimed to offer a traffic congestion prediction model that will help to predict the traffic congestion of a particular area in a definite time period. During working with machine learning models or algorithms there is a concern about the accuracy of the result. To overcome this problem, 5 different machine learning models which are used decision tree, random forest, logistic regression, SVM, and MLP to predict the congestion rate. The authors compared those models with each other and calculated the mean absolute error for each of the models so that the prediction can be more accurate. Efforts are made to alleviate the traffic congestion reducing commute times and lower carbon emissions and to enhance the overall quality of life in cities.**

Keywords—traffic congestion, decision tree, random forest, logistic regression, time series analysis.

I. Introduction

The relentless expansion of the urban landscape has brought with it a growing challenge that reverberates across metropolitan corridors – the pervasive and persistent problem of traffic congestion. The consequences of this challenge are keenly felt by city dwellers, with an average monthly global decrease of 2% in vehicle speeds due to the ubiquitous nature of traffic congestion. This not only hampers daily commuting but also poses a serious impediment to the seamless orchestration of modern urban planning and transport management. It also wastes a huge amount of man-hours [1].

In response to the urgent need for effective solutions, this study conducts a comprehensive investigation into the field of traffic jam forecasting, an important linchpin in the arsenal of tools required for contemporary urban planning strategies. The overarching goal is to proactively anticipate and mitigate congestion-related challenges, thereby encouraging the development of smarter and more adaptive transportation systems that match the demands of our dynamic urban environment.

Central to this research is the application of machine learning models, an arsenal of computational techniques adept at discerning patterns and extracting insights from complex datasets. The models listed for validation in this study cover a range of models, including decision trees, random forests, logistic regression, support vector machines (SVMs), and multi-layer perceptron (MLPs). Each model brings to the fore a unique set of characteristics and capabilities, which contribute to the collective effort to predict and subsequently address congestion.

The authors of this paper live in Dhaka, Bangladesh which has the slowest traffic speed in the world. The average speed of vehicles in Dhaka dropped from 21 km per hour in 2007 to a meager 4.8 km per hour in 2022. This speed is slightly higher than the speed of human walking [2]. Which is a huge concern for the country. Because of this problem Bangladesh loses a huge amount of working hours. Which pulls back a country economically. This is what motivated the authors to work with traffic congestion prediction so that they can find a solution to reduce this traffic problem.

The focus of this paper is how machine learning, with its multitude of algorithms, can be used as a powerful tool in congestion prediction and mitigation. Rigorous evaluation criteria, primarily anchored in the mean absolute error metric, provide a quantitative measure of each model's predictive ability. By identifying the most parsimonious model that fits the nuances of our dataset, this study aims to contribute not only empirical findings but also nuanced insights to the ongoing discourse on intelligent transportation systems. It seeks to lay the foundation for the formulation and implementation of more effective urban planning and traffic management strategies to suit the evolving dynamics of our urban landscape.

II. Related Works

To decrease traffic congestion, this study focuses on the strategic use of data mining techniques and predictive

models. Many research has been done on this topic using various models previously.

An overview and analysis of different methodologies and approaches used in traffic congestion prediction. Traffic management and planning depend on accurate congestion forecasts. It also categorizes traffic congestion prediction methods [3]. Three machine learning techniques were presented in this paper for predicting short-term traffic congestion. A real-time online model and an offline model were developed based on historical data. It has been shown that congestion can be predicted accurately in partially connected traffic streams [4]. A machine learning classification algorithm - random forest is used to construct the traffic congestion state prediction model. Random forest algorithms have high robustness, performance, and practicality. Also provides an effective method for traffic congestion forecasting and a scientific basis for management to prevent traffic congestion [5]. A new framework was build named MF-TCPV by using LSTM-SPRVM. This MF-TCPV gives a better visualization than the previous frameworks [6]. To analyze the traffic and predict the congestion on specific paths and notify vehicles intending to travel on congested paths well in advance. The logistic regression model gave better accuracy and results after applying different models [7]. A machine learning approach was presented to generate vehicle data through GPS and apply the Gaussian process to predict traffic speed. In real-time and future traffic prediction, machine learning can provide live traffic predictions in real-time [8]. Four deep learning models and three conventional machine learning models were used and Deep learning models performed better. Among deep neural networks, RNN, LSTM, and GRU performed similarly. In long-term prediction, GRU and LSTM perform better than RNN. It is clear from the CNN model that it can learn both temporal and spatial characteristics of traffic flow space-time images [9]. A new traffic jam prediction model based on pre-calculated density from the node information table. Their circular model can be mapped into urban traffic networks using road agents for traffic congestion forecasting. Using these models, traffic density tables are created after a specific time interval for each lane at each intersection [10]. A Modeling Smart Road Traffic Congestion Control using Artificial Back Propagation Neural Networks (MSR2C-ABPNN) was proposed. The goal was to provide a solution that will increase the comfort level of travelers to make intelligent and better transportation decisions [11]. The Ising-model-based method breaks away from the trade-off leveraging the Ising model's strong expressivity and the Ising machine's strong computation power. Also, Ising-Traffic formulates traffc prediction under uncertainty into two Ising models: Reconstruct-Ising and Predict-Ising. Reconstruct-Ising is mapped onto modern Ising machines and handles uncertainty in traffic accurately with negligible latency and energy consumption, while Predict-Ising is mapped onto traditional processors and predicts future congestion precisely with only at most 1.8% computational demands of existing solutions [12]. A comprehensive survey on deep learning-based approaches in traffic prediction from multiple perspectives was provided. Traffic prediction plays a crucial role in intelligent transportation systems. Accurate traffic prediction can assist in route planning, guide vehicle dispatching, and mitigate traffic congestion. This paper presents a comprehensive survey on deep learning-based approaches in traffic prediction. It summarizes existing traffic prediction methods, lists state-of-the-art approaches in different traffic prediction applications, and provides widely used public datasets in the existing literature. The paper also evaluates and analyzes the performance of different methods on a real-world public dataset [13]. Investigates different models for traffic accident risk prediction, ranging from old statistical models to new models inspired by the advent of machine learning. It presents a comparative study of these models to analyze and deduce a beneficial approach to traffic accident risk prediction [14]. A prediction model for traffic congestion that utilizes day, time, and weather data, such as temperature and humidity was proposed. The model has been tested against traffic data from New Delhi and can predict congestion one week ahead with an average RMSE of 1.12 [15]. Deals with traffic prediction that can be done in intelligent transportation systems which involve the prediction between the previous year's data set and the recent year data which ultimately provides the accuracy and mean square error. This prediction will be helpful for the people who are in need to check the immediate traffic state [16]. Applies some of the common and familiar machine learning concepts like Deep Auto encoder (DAN), Deep Belief Network (DBN), and Random Forest (RF) on the online dataset for the traffic flow predictions [17]. S. K. Banshal did a thorough investigation with themes and trends in CS [18]. M. Abdullah-Al-Kafi shows the relationship between human and Ai in ML [19], has also use ML for fake news classification [20]. A multi-class support vector machine method was proposed for pattern recognition to intelligently judge traffic congestion in intelligent traffic. It aims to improve data processing and real-time traffic state identification for road congestion [21]. Traffic congestion was analyzed in the main streets of Electronic City using a traffic congestion index and artificial neural network [22]. [23]. A study on quantifying traffic congestion on an urban arterial road in Thiruvananthapuram city was presented. Congestion cost and index was evaluated to prioritize traffic management measures effectively. Congestion indices under heterogeneous traffic conditions in Bhubaneswar city was evaluated. And also the impact of lane discipline was analyzed on congestion indices and focuses on travel time reliability measures[24].

III. METHODOLOY

The purpose of this study is to find the most effective model among some selected models to predict sales. This will improve the efficiency and reliability of forecasting sales.

A. Data Collection and Preparation

In order to obtain traffic information, first identified a reliable source. In this study, the authors utilized the "Traffic Prediction Dataset" fedesoriano. (February 2021). Traffic Prediction Dataset. Retrieved [Date Retrieved] from https://www.kaggle.com/datasets/fedesoriano/traffic-prediction-dataset, a data set that provides historical traffic records. This selected data was acquired from the Kaggle dataset and ensured that it covered a relevant time period and included features like Junction, Vehicles, and ID. A total of 48120 data points are presented in each of the four columns of this dataset.

B. *Data Preprocessing*

Preprocessing data before analyzing it is the most important step for getting accurate results. There was four different attributes in data: ‘DateTime’, ‘Junction’, ‘Vehicles’, and ‘ID’. However, ID was not important for predicting traffic. Therefore, the attribute ID was removed from our dataset. In order to identify missing values, outliers, and inconsistencies in the acquired data, a thorough quality check was conducted. The process of data cleaning involves imputation of missing values (e.g., mean or median imputation) or the removal of instances with missing data if they represent a small fraction of the data. This dataset did not contain any missing values. Therefore, it was not need to go through this process.

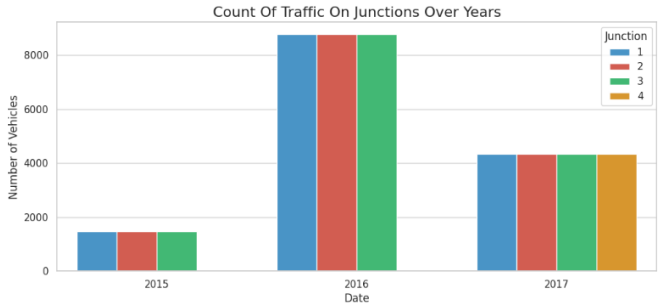


Fig. 1. Traffic on Junctions over Years

After completing the process of outlier detection, outliers were found in “Item Visibility” and “Item Outlet Sales” data columns as shown in Fig. 1.

C. *Feature Engineering*

C.1 Feature Identification and New Feature Creation

A Date-Time conversion was performed on the column "DateTime" using pandas.to_datetime(), which converts arguments to date times. Several additional features have been added based on the time and date information in the "DateTime" column, such as "Year", "Month", "Date", "Hour", and "Day". Since Day was created in no-numerical form, it had to be converted to its numerical form. In this process, categorical data is handled. Categorical variables can be converted into a machine-readable format for machine learning. Converted 'Day' into numerical format using sklearn.preprocessing library's Label Encoder method.

C.2 Feature Selection

A feature selection technique was applied to identify the most relevant features and eliminate redundant or irrelevant ones. A correlation matrix was computed for the dataset. These coefficients reveal the strength and direction of linear relationships between all pairs of numerical variables.

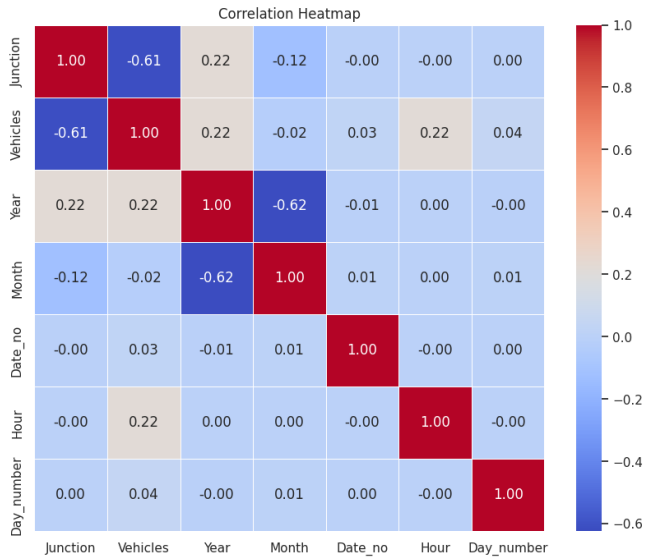


Fig. 2. Heat map for Correlation between attributes

The heat map shown in Fig.-02 was created using a data visualization library such as Seaborn. The figure size, color map, and other parameters were adjusted to enhance the clarity and interpretability of the heat map. Heat maps were used to inform feature selection. These relationships enable to select features for further analysis based on redundant information provided by highly correlated features. The model includes 'Year', 'Month', 'Date_no', 'Hour', and 'Day_number' as independent variables (x) and 'Vehicles' as a dependent variable (y).

D. *Model Selection and Training*

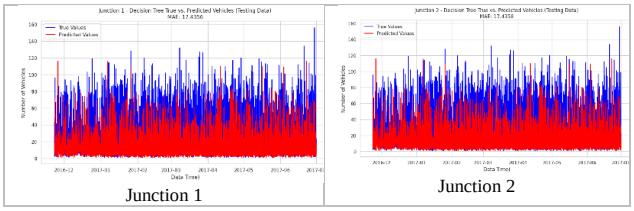
D.1 Data Splitting

As part of the machine learning workflow, data splitting is a crucial step, in which the dataset is divided into training, validation, and test sets. Through this process, it was possible to assess how well the machine learning model performs on new, unseen data. Using the preprocessed data, 70-80% of the data were used for training while the remaining 20-30% were used for testing.

D.2 Algorithm Selection

In order to capture the temporal patterns and relationships between variables, machine learning algorithms were selected which are suitable for capturing such patterns and relationships. These algorithms are Decision Trees, Random Forests, Logistic Regression, Support Vector Machines, and MLP. For each of these model time series analysis was used to show the difference between the predicted and true value.

1) Decision Tree: Decision Tree is a type of nonlinear tree-based model. Predictions are made by recursively splitting data by features. Classification and regression tasks can be performed using these.



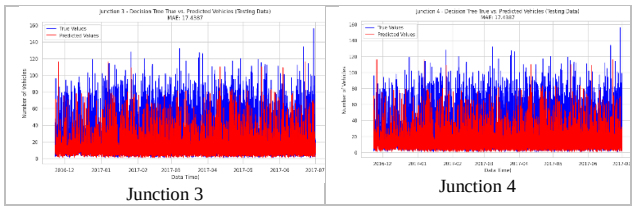


Fig. 3. Decision Tree predicted vs True vehicles.

Fig. 3 shows the predicted value of vehicles for each date and time of the dataset using Decision Trees.

TABLE I. MEAN ABSOLUTE ERROR OF DECISION TREE

Junctions	Mean Absolute Error
Junction 1	17.4356
Junction 2	17.4389
Junction 3	17.4368
Junction 4	17.4343

Based on Decision Tree, MAE (Mean absolute error) values for each junction are shown in TABLE I.

2) Random Forest: A Random Forest is a machine learning algorithm that uses ensemble learning to solve classification and regression problems. The output of many decision trees is combined to produce a single result using this supervised learning algorithm.

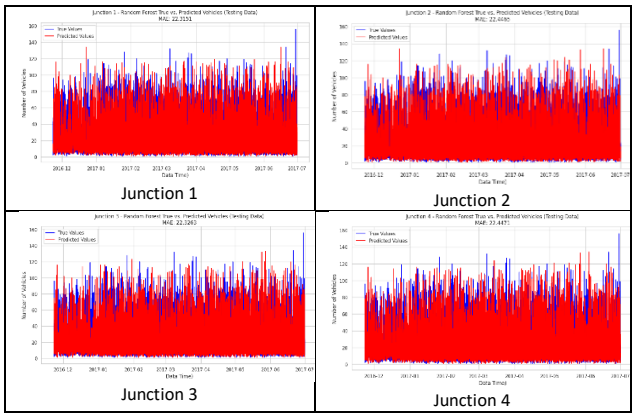


Fig. 4. Random Forest predicted vs True vehicles.

In Fig. 4 the predicted value of vehicles each date and time of the dataset was shown using Random Forest for all the junctions.

TABLE II. MEAN ABSOLUTE ERROR OF RANDOM FOREST

Junctions	Mean Absolute Error
Junction 1	22.3151
Junction 2	22.4465
Junction 3	22.3263
Junction 4	22.4471

According to TABLE II, MAE (Mean absolute error) is displayed for each junction based on Random Forest.

3) Logistic Regression: Logistic Regression is a statistical model commonly used for binary classification tasks, where the outcome variable is categorical and has two classes. Despite its name, logistic regression is primarily applied for classification rather than regression.

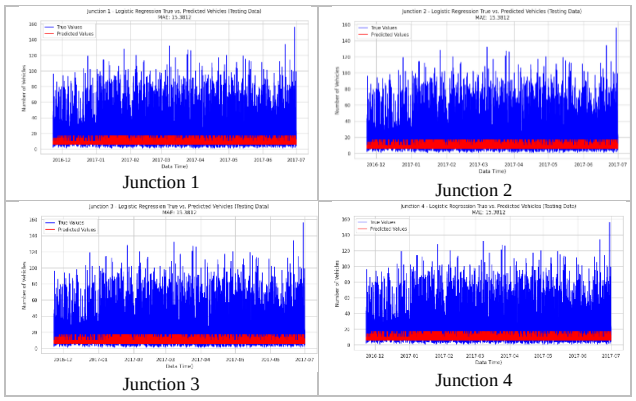


Fig. 5. Logistic Regression predicted vs True vehicles.

In Fig. 5, the predicted value of vehicles each date and time of the dataset was shown using Logistic Regression for all the junctions.

TABLE III. MEAN ABSOLUTE ERROR OF LOGISTIC REGRESSION

Junctions	Mean Absolute Error
Junction 1	15.3812
Junction 2	15.3812
Junction 3	15.3812
Junction 4	15.3812

In TABLE III the MAE (Mean absolute error) value of each junction using Logistic Regression was shown.

4) Support Vector Machine (SVM): As a machine learning algorithm, support vector machines (SVMs) solve classification, regression, and outlier detection problems by using supervised learning models. SVMs are particularly effective in solving binary classification problems, which require classifying the elements of a dataset into two groups.

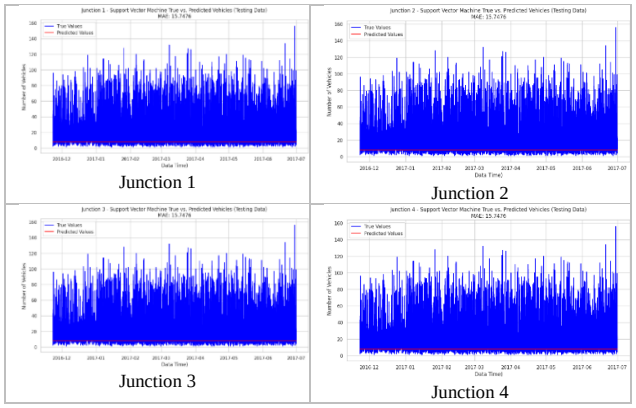


Fig. 6. SVM predicted Vehicles vs True vehicles.

A SVM (Support Vector Machine) was used to predict vehicle values at each junction for all the dates and times of the dataset in Fig. 6.

TABLE IV. MEAN ABSOLUTE ERROR OF SVM

Junctions	Mean Absolute Error
Junction 1	15.7476
Junction 2	15.7476
Junction 3	15.7446
Junction 4	15.7476

In TABLE IV the MAE (Mean absolute error) value of each junction using SVM (Support Vector Machine) was shown.

5) A Multilayer Perceptron (MLP): MLP is an artificial neural network (ANN). It is a fully connected neural network with three layers: input, hidden, and output layers. Each layer has multiple connected nodes.

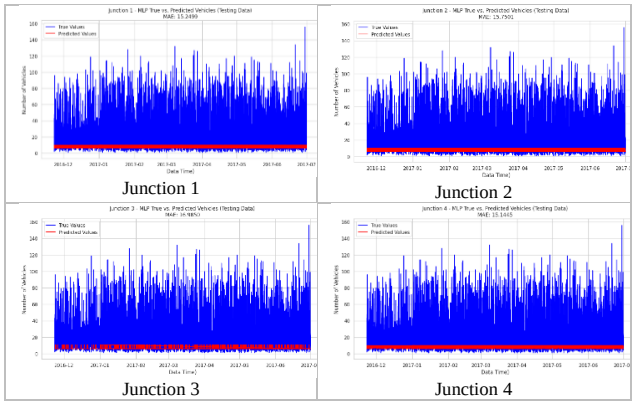


Fig. 7. MLP predicted vs True vehicles.

In Fig. 7. the predicted value of vehicles each date and time of the dataset was shown using MLP for all the junctions.

TABLE V. MEAN ABSOLUTE ERROR OF MLP

Junctions	Mean Absolute Error
Junction 1	15.2499
Junction 2	15.7501
Junction 3	16.9850
Junction 4	15.1445

In TABLE V the MAE (Mean absolute error) value of each junction using MLP was shown.

In evaluating these models, MAE values of each model were calculated, and then compared them to identify the best model. Based on these graphical representations, it was possible to compare how well each model aligned with actual traffic conditions.

IV. RESULT ANALYSIS

The study systematically evaluated the performance of five traffic congestion prediction models: Decision Tree, Random Forest, Logistic Regression, Support Vector Machines (SVM), and Multi-Layer Perceptron (MLP), using the mean absolute error (MAE) as a metric. To understand the accuracy of each model, a time series analysis was conducted, and the results were visualized in Fig. 8. The matplotlib bar graph plot enables a direct comparison of the models' performance based on their respective MAE values.

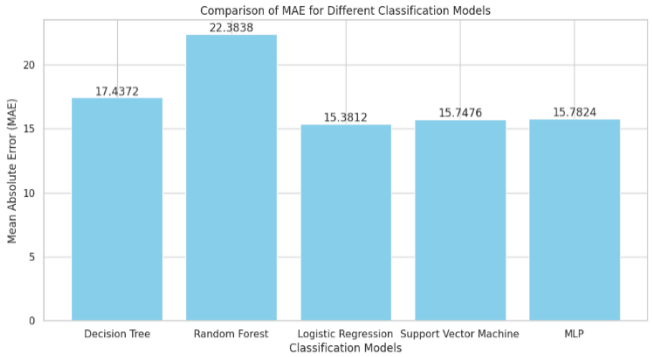


Fig. 8. Comparison of MAE of all five Models

Fig. 8 illustrates scenarios where models performed sub optimally, providing a foundation for error analysis. Valuable insights into prediction error factors were gained, guiding potential model enhancements and improvements in data preprocessing. To get the most accurate results, we must have the lowest mean absolute error.

TABLE VI. MEAN ABSOLUTE ERROR OF EACH MODEL

Models	Mean Absolute Error
Decision Tree	17.4372
Random Forest	22.3838
Logistic Regression	15.3812
Support Vector Machine(SVM)	15.7476
MLP	15.7824

Table VI presents the MAE values for each model. Logistic Regression emerged as the most accurate model, boasting the lowest MAE (15.3812). The dataset supports this conclusion, suggesting high accuracy in predicting traffic congestion.

Despite its ensemble nature, Random Forest showed a higher MAE than Decision Tree. The study recommends a further investigation into Random Forest's performance, which may involve tuning hyperparameters or exploring feature importance to refine its predictive capabilities. SVM and MLP exhibited competitive performance, positioning them as suitable models for traffic congestion prediction. The study suggests additional improvements through fine-tuning these models or experimenting with different architectures. Logistic Regression stood out as the most useful model for predicting traffic congestion, attributed to its consistently low MAE. However, the study acknowledges the potential for performance improvement through ongoing experimentation and fine-tuning.

The findings underscore the significance of ongoing model refinement for traffic congestion prediction. The study concludes that Logistic Regression, while effective, can benefit from continuous improvement efforts. Future research may focus on refining models, exploring alternative architectures, and adapting to evolving urban dynamics. The study acknowledges limitations, such as the need for more granular data and external factors influencing traffic patterns. Recommendations include expanding datasets and exploring advanced model architectures for improved accuracy.

In conclusion, the study provides valuable insights into the performance of traffic congestion prediction models. Logistic Regression stands out as the most accurate, but ongoing experimentation and fine-tuning remain crucial for enhancing predictive capabilities and addressing urban traffic challenges.

V. Conclusion

To control traffic congestion, this study utilized a variety of machine learning algorithms, such as decision trees, random forests, logistic regression, and support vector machines. A classical but robust model, logistic regression, was specifically tested for its efficiency in addressing traffic congestion. Logistic regression showed promising results despite its simplicity. Despite its simplicity, logistic regression offers a balance between effectiveness and complexity. It provides insights for policymakers aiming to improve urban transportation systems as well as academic discussions. Traffic control strategies can be improved by integrating machine learning techniques, especially logistic regression, to alleviate congestion. Data-driven decision-making approaches, including logistic regression, will be key to improving traffic management strategies in the future.

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